

Experiment No- 6

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Subject:- Data Analytics

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6) Analyzing the Value Distribution of Arrays Generated by np.random.randint().

1) randint():-

```
import numpy as np

# Generate a random integer between 1 and 10 (inclusive)
random_int = np.random.randint(2, 21)
print(random_int)
```

□ Output:-



20

```
import numpy as np

# Generate a 2x3 array of random integers between 10 and 20
random_numbers= np.random.randint(10, 21, size=6)

print("Random numbers:",random_numbers)
```

□ Output:-

```
Random numbers: [10 19 16 20 20 15]
```

2) rand():-

```
import numpy as np # Import the NumPy library

# Generate a single random float in the range [0.0, 1.0)

random_float = np.random.rand()

# Print the generated random float

print(random_float)
```

□ Output:-

```
0.6255187867159914
```

```
import numpy as np # Import the NumPy library

# Generate a 2x3 matrix of random floats between 0.0 and 1.0

random_matrix_2d = np.random.rand(2, 3) # Shape: 2 rows, 3 columns

# Print the generated 2D random matrix

print(random_matrix_2d)
```

□ Output:-

```
[[0.74928822 0.16820477 0.36122282]  
 [0.18498579 0.46393551 0.83063499]]
```

3) randn():-

```
import numpy as np # Import the NumPy library  
  
# Generate a single random float from the standard normal distribution  
  
# (mean=0, standard deviation=1)  
  
random_float = np.random.randn()  
  
# Print the generated random float  
  
print(random_float)
```

□ Output:-

```
0.5611617145544704
```

```
import numpy as np # Import the NumPy library

# Generate a 1D array of 5 random floats from the standard normal distribution
one_d_array = np.random.randn(5)
# 5 samples from N(0, 1)

# Print the generated 1D array
print(one_d_array)
```

□ Output:-

```
[ 0.91246482 -0.35928503 -0.3798294 -0.0063953  0.91113271]
```

4) ranf():-

```
import numpy as np

# Generate a single random float
single_random_float = np.random.ranf()

print(f"Single random float: {single_random_float}")

# Generate an array of 5 random floats
random_array = np.random.ranf(5)

print(f"Array of 5 random floats: {random_array}")

# Generate a 2x3 array of random floats
random_2d_array = np.random.ranf((2, 3))
```

```
print(f"2x3 array of random floats:\n{random_2d_array}")
```

- **Output:-**

```
Single random float: 0.34338383835097397
Array of 5 random floats: [0.24835924 0.49597631 0.20252274 0.15274063 0.46522801]
2x3 array of random floats:
[[0.63738679 0.90208278 0.20974219]
 [0.21736777 0.80536427 0.07966398]]
```

```
import numpy as np
```

```
# Legacy method
```

```
random_array_1d = np.random.ranf(5)
```

```
print(f"Legacy ranf(5): {random_array_1d}")
```

```
# Modern method
```

```
rng = np.random.default_rng()
```

```
random_array_1d_modern = rng.random(size=5)
```

```
print(f"Modern random(size=5): {random_array_1d_modern}")
```

- **Output:-**

```
Legacy ranf(5): [0.17937469 0.35840866 0.16669717 0.75624744 0.24474419]
Modern random(size=5): [0.19612123 0.88031426 0.60386923 0.67894355 0.64992251]
```
